NATHACKS 2026 · CONSTRUCTION-SAFETY WEARABLE

ThunderClock2026 system report: simulator, physics viewer, prediction model and human model

One document for the four large pieces built so far: the ThunderSim simulator, the 3D viewer with its physics engine, the fall and collapse prediction model with its proposed detection architecture, and the human models (simulated body, viewer rig and ragdoll, and BodyScan).

Date	Repository	Status
7 November 2026	nathacks/ThunderClock2026, GitHub main 656b209	Prototype, simulated data only

1 The system at a glance

A four-sensor strip runs down the worker's spine (pelvis, lumbar, thoracic and upper thoracic). An ESP32-S3 reads the sensors through an I2C multiplexer and streams over Bluetooth Low Energy to a Python backend on a laptop, which runs the detector and feeds the supervisor dashboard. Everything in this report was built to test and show that chain before real falls are ever recorded.

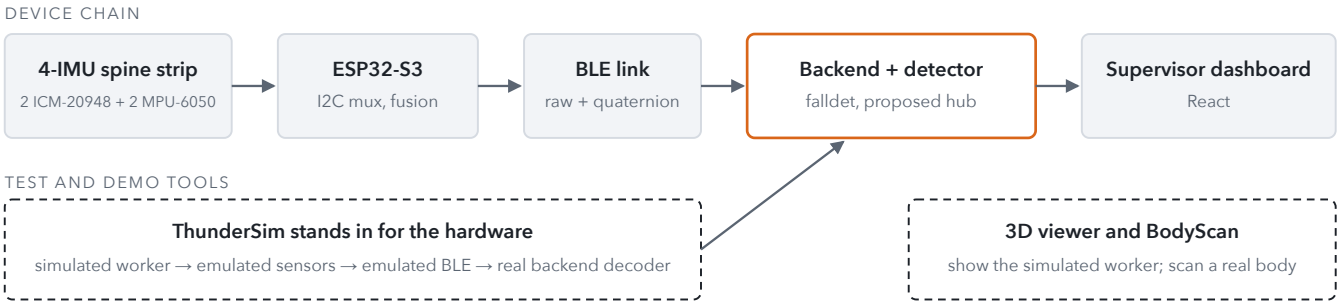


Figure 1. The device chain (top) and the tools built around it (bottom). The detector box is where the prediction model runs.

ThunderSim simulator

`detection/simulation/thundersim`

Simulates a construction worker wearing the strip through 35 scenarios, emulates the sensors and the radio link, decodes packets with the real backend code, and scores any detector.

3D viewer with physics

`detection/simulation/thundersim/viz`

Replays any run in a browser: the worker, the strip and the site. A numpy physics engine now drives the limbs, the lanyard, the safety net and dropped objects.

Prediction model (falldet)

`detection/models` · `docs/DETECTION_ARCHITECTURE.md`

Two-stage fall detector: a threshold trigger that ports to the ESP32, then a gradient-boosted classifier over the 8 s after it. A detection architecture adds collapse and man-down paths.

Human models

`thundersim/spine.py` · `viz/rig.py` · `viz/physics.py` · `bodyscan/`

A 4-link spine that produces the sensor data, a procedural 18-joint rig with a ragdoll for display, and BodyScan, which fits a 3D body to four phone photos.

PIECE	STATE	VERIFIED BY
ThunderSim	WORKING	86 automated tests pass (unittest, 98 s)
3D viewer and physics	WORKING	Sensor data byte-identical to the original viewer in 24 checked runs; browser checks across 14 scenarios
falldet	WORKING, WRONG MODEL PICKED	Leave-one-subject-out on lab data; ThunderSim runs on 20 dev seeds
Collapse and man-down hub	DESIGNED ONLY	Prototype probes; `detection/hub/` not built
BodyScan	WORKING ON TEST INPUTS	Server tests with random weights and simulated photos; real CameraHMR weights not yet run
Firmware streaming	BLOCKER	GitHub main cannot deliver 100 Hz raw data yet (section 4.5)

COMPONENT 1

2 ThunderSim, the fall and man-down simulator

ThunderSim tests detectors before anyone wears the strip. It generates physically consistent data for the four spine sensors on a simulated worker, sends it through the same path real packets take, and scores the detector on falls, near-falls, construction activities that look like falls, and full fall-free shifts. It is plain numpy and scipy, with no game engine.

What one run does

STEP	WHAT HAPPENS
Scenario script	Picks a worker (height, mass, clothing), a site (surface, wet or icy, temperature, radio) and an event, randomised by seed.
Body model, 1 kHz	Moves a 4-link spine through the activity; during falls, physics takes over (section 5.1).
Soft tissue and strap	A second-order mount filter per sensor (8–25 Hz, damping 0.4–0.9), softer in winter clothing.
Sensor emulator	MPU-6050 low-pass filter table, noise, bias, temperature drift, int16 rounding, clipping, and mux reads 0.8–1.5 ms apart sharing one timestamp.
BLE link	Packet loss, outages and throughput limits, including a case where 100 Hz raw streaming overflows a 6 kB/s link.
Backend	Decodes the protobuf with the team's unchanged `backend/app/protocol.py`, then calibrates to a neutral pose.
Detector and scoring	Any model with `reset`/`update` returning events; then worker check-in, supervisor alerts, and the pass bars of the simulation spec.

Physics of falls (from `falls.py` and `site.py`)

- **Loss of balance:** inverted pendulum over the stance foot with knee collapse. **Slips:** the foot slides out, then slows by kinetic friction; a slip happens when the slide distance $v^2/(2\mu g)$ exceeds 0.15 m.
- **Flight:** ballistic about the centre of mass, so a tumbling body reads $\omega^2 r$, not 0 g. Free-fall time matches $\sqrt{(2h/g)}$ (tested).
- **Body-first impact:** effective mass 0.35 × body mass and pelvis stiffness 39–55 kN/m in series with the surface. Hip contact speeds come out at 3.3–4.0 m/s against the published 3.4 ± 0.5 m/s (tested).
- **Harness arrest:** OSHA limits of at most 1.8 m free fall, a stop within 1.07 m and at most 8 kN (tested), then hanging 15–35° forward with pendulum swing. Safety nets sit at most 9.1 m below the work level.
- **Site:** ten surfaces from concrete to safety net, each with stiffness, damping, boot friction and unevenness; ladders at 1:4.

Scenarios

GROUP	SCENARIOS (35 IN TOTAL)
Falls from height	H-LAD, H-SCAF, H-ROOF, H-HOLE, H-EXC
Same-level falls	L-SLIP, L-TRIP, STAIR, STRUCK-KD
Arrested falls	ARR (harness), NET
Collapse	SYNC, SYNC-KNEEL (report only)
Near-falls	NF-STUMBLE, NF-RAIL, STRUCK-NF, SLIP-FRICTION
Look-alikes (hard negatives)	Hammering, overhead drilling, jackhammer, equipment cab, kneeling and crawling, rebar tying, ladder, jump-down, hard sit, lying down, declared lying work, lunch nap, carrying, running, hard hat knocked off, strap slip, strip left on a table

Shifts mix these activities in the spec's proportions, plus about 10 jump-downs and 5 runs per 8 h, to measure false alarms per shift.

Firmware profiles and the calibration finding

PROFILE	RATE	ACCEL	GYRO	FILTER	MEANING
`current`	0.5 Hz	±2 g	±250 °/s	10 Hz	Original firmware
`current_100hz`	100 Hz	±2 g	±250 °/s	10 Hz	Original with the sampling period fixed
`target`	100 Hz	±16 g	±2000 °/s	44 Hz	Research recommendation
`fw_nov7` (probe)	100 Hz	±8 g	±1000 °/s	≈10 Hz	Firmware settings on GitHub main today

The `calibrate` command compares simulated standing falls onto mats with the Erciyes lab falls. The lab median peak is 3.95 g (IQR 3.3–4.6 g) against 6.5–7 g simulated at the published hip speed. Soft tissue cannot explain the gap, since the mount filter passes everything below 12.5 Hz. The volunteers made controlled falls onto thick mats, so site falls are deliberately **not** scaled down: a model trained on lab data will see larger impacts on site.

> Limits: the body is kinematic with physics only in fall phases; many design values are untuned against real recordings; real worker falls are not simulated in any validated sense. Results are a necessary gate before pilot data, not validated performance.

COMPONENT 2

3 The 3D viewer and its physics engine

The viewer replays any ThunderSim run in a browser with three.js: the worker, the strip, the site props and the sensor charts on a timeline. The original viewer posed the limbs procedurally and played falls in slow motion by default, which made gravity look about 16 times too weak. The new version adds a physics engine, rewrites the limb animation and fixes the props. **It changes only what is drawn**: the simulated sensor signals are untouched.

3.1 The physics engine (`viz/physics.py`, thundersim-pbd v1.1)

- **Method:** extended position-based dynamics (XPBD) written in numpy, deterministic, 250 Hz substeps (5 per 50 Hz frame). Each substep predicts positions under gravity, applies muscle drives and pins, solves distance constraints, bend and cone joint limits and a centre-of-mass constraint, then resolves collisions with friction against boxes, capsules, sheets and the floor.
- **Ragdoll:** hips, shoulders and neck ride on the simulated spine; knees, ankles, heels, toes, elbows, hands and head are free particles. A muscle-tone schedule comes from the simulator's ground-truth events: full tone while standing, limp after an impact (for example 0.08 when unconscious), and back to full tone over 0.8 s for a struggle or recovery, with drives critically damped so limbs do not overshoot.

- **Free flight:** while falling, the limbs keep the drawn body's centre of mass on the simulator's ballistic path, so the whole figure falls at g.
- **Props on the same engine:** a 13-point lanyard rope with an energy-absorbing pay-out of up to 1.07 m; a safety-net cloth (0.29 m grid, 8 % pretension) that the body dents; a hard hat as six rigid points on a 0.55 m cable. The crane load is a single 10 m pendulum timed so the beam reaches the bottom of its arc at the struck moment.
- **Where it runs:** only in the time windows that need it. A final pass restores exact bone lengths, and a display-only lift of at most 0.15 m keeps a resting trunk out of the floor.
- **Playback:** real time by default; slow motion is opt-in (0.25× from 0.5 s before to 1.3 s after each fall event, eased, with an on-screen badge). The camera follows height through a critically damped 0.8 s spring.

A ready-made engine (Jolt) was considered, but running downloaded external code was not allowed in this session, so the engine is custom. It costs 1.12–1.39× the build time of a physics-off payload for a 60 s run.

3.2 Results, original viewer against the new one

CHECK	ORIGINAL	NEW, PHYSICS ON	RUNS
Gravity seen on screen during falls	0.61 m/s²	**9.81 m/s²**	71 of 71 flights
Feet jumping more than 30 cm in one frame	503	**0**	105 runs (35 scenarios × 3 seeds, 20 s)
Knee or elbow flips	173	**4, all at impacts**	
Foot jumps over 12 cm per frame, outside falls	710	**3**	
Head turns over 20° per frame, outside falls	0	0 (worst 16°)	
Hand jumps over 30 cm per frame	42	**0**	60 s set
Resting torso height (median, body-first falls)	−5.8 cm	**+0.3 cm**	42 falls
Share of resting frames more than 2 cm in the floor	80 %	**2 %**	
Worst head below the floor	−36 cm	**−1.2 cm**	
Body below the net cloth during a catch	0.57 m	**≤ 4 mm**	NET seeds
Lanyard stretched past its length	up to 6×	**≤ 0.5 %**	ARR seeds

3.3 How it was verified

- **Sensor data unchanged:** in 24 runs (12 scenarios × seeds 0 and 3, 60 s), signals, segment and chip orientations, phases, worn flag, ground truth, results and events are byte-identical to the original viewer with physics on and off. With physics off the spine and strip positions are also identical; with physics on they differ only by one shared vertical lift per frame (largest 0.149 m, in 14 of 24 runs), which is noted in the payload.
- **Process:** four build agents, then a verification stage (metrics, headless-browser screenshots, invariants, code review) that found 14 problems (6 high, 8 medium), then a fix stage that resolved them; 86 tests pass.

> Open: in mid-air, hands accelerate about twice as hard with physics on (median 115 against 61 m/s²) where limbs scrape trench walls, roof edges and scaffold. Some procedural lying and kneeling poses put knees 5–7 cm below the floor outside physics windows. The simulator itself makes bodies bounce 0.5–1.26 m after hard impacts, the net throws the worker back up 1–2.4 m, and trips topple at about 1.3–1.7 g; these are ground truth and are shown as simulated.

COMPONENT 3

4 The prediction model and the detection architecture

`falldet` (owned by research-95, in `detection/models/`) outputs `normal`, `near_fall` or `fall_confirmed` with a confidence. It is trained on public lab data; every training fall was acted by young volunteers onto mats, so real-world performance is expected to be lower.

4.1 How it works

STAGE	RULE OR MODEL
Stage 1 trigger	Per sample, `svm_g` = mean pelvis/lumbar acceleration in g and `rot` = max pelvis/lumbar rate. Fires on an **impact** (> 2.25 g; 1.9 g for the 10 Hz model) plus free fall (< 0.7 g for ≥ 100 ms) or rotation (> 3 rad/s) within 1 s; 5 s refractory. Thresholds chosen by grid search inside every training fold. Written to port to ESP32 C++.
Stage 2 confirmer	Gradient-boosted trees (depth 3, 200 iterations) over 2 s before to 8 s after the trigger, 28 features: free-fall dip, impact peak, jerk, rotation, activity before and after, whether walking resumed, and four-sensor features (median inclination, how many sensors lie past 60°, spread, stillness).
Post-rules	Harness arrest: ≥ 0.35 s free fall then no walking confirms; ≥ 0.6 s always confirms. Lying-work mode downgrades a fall if the worker was already lying.

4.2 Lab results (leave-one-subject-out, 30 folds, Erciyes and VTT-ConIot)

FIRMWARE SETTING	SENSITIVITY	SPECIFICITY	F1	FALSE ALARMS/H	STAGE 1 RECALL
Target (±16 g, ±2000 °/s, 44 Hz), shipped model	0.969	0.995	0.981	0.73	0.972
Current (±2 g, ±250 °/s, 10 Hz), retrained	0.966	0.995	0.980	0.64	0.968
Current data, target-trained model	0.738	0.999	0.849	0.09	0.745

Falls ending on the floor are confirmed in 93–100 % of recordings and falls with recovery are labelled near-fall in 98–100 %. Weak spots: stumbles without a hard impact (Stage 1 fires on 53 %) and fainting while sliding down a wall. Construction data (3.3 h) gave no false confirmed falls. Lab false alarms came from lying down on a bed and sitting hard on a sofa.

4.3 Results in ThunderSim on today's firmware settings (`fw_nov7`, 100 Hz delivery, 20 dev seeds)

CATEGORY	FALLDET AS SHIPPED	FALLDET FORCED TO 10 HZ MODEL	RULE BASELINE
Falls from height	98 %	not recorded	not recorded
Arrested falls	98 %	not recorded	not recorded
Same-level falls	94 %	not recorded	not recorded
Collapse confirmed	25 %	78 %	75 %
False alarms on a hard sit	not recorded	50 %	65 %

Collapse is low as shipped because `falldet` picks its model from the accelerometer range (above 4 g → the 44 Hz model) rather than from the filter setting; every miss was Stage 1 never firing. Choosing the model by filter bandwidth fixes most of it. That is research-95's decision. All figures are on dev seeds; there is no clean held-out result yet.

4.4 Collapse and the proposed architecture (`docs/DETECTION_ARCHITECTURE.md`)

- **Main model on the raw sensor queue:** cheap triggers run on every sample: T1 impact, T2 stumble, and a new **T3 topple** (upright to down while turning at ≥ 3 rad/s, no impact needed). A trigger opens a window for classification; the new **C3 collapse confirmer** checks the worker stays down and mostly still, tolerating brief movement.
- **T4 man-down timer**, armed only by a trigger: warning at 30 s, critical at 60 s; no motion after a trigger escalates instead of reading as "strip removed".
- **Arbiter and incident layer:** triggers within 10 s merge into one alert; dismissals cover one alert and never cancel man-down; a lost link during a suspected fall escalates as DATA_LOST.
- **Prototype probes:** T3 fired on 17 of 20 SYNC-KNEEL runs; the first C3 rejected 8 of 8 collapses with brief twitches (redesigned); collapse ending seated or against a wall is **not covered** yet.

4.5 Blockers before the device can detect anything

> **Firmware raw FIFO (critical path).** On GitHub main 656b209, sensor data passes through depth-1 mailboxes that keep only the latest value, with 1–2 s debug periods, and the IMU and BLE tasks are switched off. Raw data needs a FIFO of at least 32 readings and one BLE notification per reading at 100 Hz. **falldet rate latch:** any short stretch below 40 Hz turns detection off for the whole session; the design recreates it once the rate holds at ≥ 90 Hz for 5 s.

COMPONENT 4

5 The human models

Three separate human models serve three jobs: one generates the sensor data, one makes the replay look like a person, and one captures a real person's body shape.

5.1 The simulated body that produces the data (`thundersim/spine.py`)

- **Chain:** four rigid links (pelvis, lumbar, thoracic, upper thoracic), built as p(i+1) = p(i) + L(i)·axis(i), with joints in the ISB spine order. Each sensor sits 4 cm behind the spine line, so rotation adds the lever-arm terms ω×(ω×r) and α×r.
- **Movement:** soft range-of-motion limits, voluntary trunk rate capped at 180 °/s, and C1-continuous joins between motions so no velocity step can look like an impact.
- **Background motion:** breathing at 0.2–0.33 Hz (about ±0.01 g), postural sway and fidget. An unconscious worker still breathes; a strip on a table does not.
- **Limits:** no arms, legs or head segments in the simulator; impacts use single-degree-of-freedom contact.

5.2 The display figure (`viz/rig.py` and the ragdoll)

- **Rig:** an 18-joint figure built around the simulated spine with two-bone inverse kinematics for arms and legs. The rewrite plans footsteps from the simulator's own heel strikes (median timing offset 140 → 33 ms), never snaps a foot between poses, and keeps the knee and elbow bend planes continuous.
- **Rig results, physics off** (105 runs): foot jumps over 12 cm per frame outside falls 702 → 26; knee and elbow flips 167 → 0; ladder frames where a foot misses its rung by more than 2 cm 33.8 % → 0 %.
- **Ragdoll:** after a fall the limbs hand over to the physics engine of section 3.1, so they rest on the floor or net instead of pointing into the air.

5.3 BodyScan, a real person's body shape (`bodyscan/`)

PART	WHAT IT DOES
Phone app	Kotlin, Compose and CameraX on a Galaxy S24 Ultra. Takes front, left, back and right photos with the camera's intrinsics, rotation and the subject's height.
Laptop server	FastAPI. Finds the person, runs CameraHMR (ViT-H) once per scan on Apple MPS (about 0.1 s per view), fuses the per-view SMPL shape parameters with a median, rebuilds an upright mesh scaled to the entered height, and measures chest, waist and hip by slicing the T-pose mesh.
Viewer	three.js page that shows the scanned pose and T-pose, measurements and per-photo fit overlays.
Transfer	USB (works on venue Wi-Fi with client isolation) or LAN; photos never leave the phone and laptop.

> Measurements are approximate (several cm in chest, waist and hip); SMPL with 10 shape parameters, gender neutral. Tests so far used random weights, a fake SMPL file and simulated photos; the real CameraHMR weights and S24 camera values are not yet verified. CameraHMR and SMPL are licensed for non-commercial research and education only and are never committed. Scan people only with consent.

A natural next step is to use a scan's height and proportions to set the simulated worker's link lengths and the viewer figure, so the replay matches the person wearing the strip.

SUPPORTING WORK

6 Agents, documents and next steps

6.1 Claude Code agents in `.claude/agents/`

AGENT	JOB	GUARD RAILS (HOOK-ENFORCED)
`collapse-detection`	Builds `detection/hub/` (T3, C3, T4, arbiter), tunes on dev seeds and reports once on a fresh held-out seed block with confidence intervals.	Writes only under `detection/hub/`; falldet changes as diffs; no seeds 1,000,000–1,999,999 (burned)
`architecture-reviewer`	Adversarial, read-only review of the detection chain against the newest GitHub code; labels claims verified or inferred and ranks findings.	Read-only git, no installs, no held-out seeds
`mocap-curator`	Brings real motion capture or construction video into the viewer's motion library.	Licence guard blocks gated datasets and video sites; viewer-only

6.2 Documents

- `detection/simulation/HOW_IT_RUNS.pdf`: how one simulation run works.
- `docs/DETECTION_ARCHITECTURE.md`: the detection design, blockers, test protocol and 12 open questions.
- `detection/simulation/README.md`: simulator and 3D viewer reference, now including the physics engine.

6.3 Next steps, in order

#	STEP	OWNER
1	Add the raw FIFO to the firmware and measure the real delivery rate on the laptop.	Firmware (team to assign)
2	Pick the falldet model by filter bandwidth and make the rate latch recoverable.	research-95
3	Run the collapse-detection agent to build the hub and get a held-out collapse result.	Detection
4	Add seated and wall-slide collapse scenarios; review the post-impact bounce, net rebound and fast topples.	Simulator owner
5	Record real strip data and run `thundersim calibrate`, then re-run the suite.	Team
6	Run BodyScan with the real CameraHMR weights and feed body dimensions into the simulator and viewer.	BodyScan

> **Wording for the demo:** prototype indicator. Say "fall", "near-fall", "collapse (no impact)" and "man down". All results are on simulated or lab-acted data; make no medical or compliance claims.